

Log-Periodic Transseries for the Inverse of a Real Fibonacci Continuation

An Explicit Lagrange–Good Calculus for Exponential–Oscillatory Inversion

Vladimir Reshetnikov
ProveIt research note

3 September 2026

Abstract

Let

$$\mathcal{F}(x) = \sqrt{\frac{2}{5}} \sqrt{\cosh(2x \operatorname{arccsch} 2) - \cos(\pi x)}, \quad x \in \mathbb{R}.$$

Because $\operatorname{arccsch} 2 = \log \varphi$, where $\varphi = (1 + \sqrt{5})/2$, this is the modulus of the real-parameter Binet continuation and satisfies $\mathcal{F}(n) = |F_n|$ at every integer. The function is even. On $[2, \infty)$ it is strictly increasing from 1 to ∞ , so it has an outer positive inverse $\Psi(y)$; the corresponding negative inverse is $-\Psi(y)$.

This article obtains an exact all-orders expansion of these inverse branches as $y \rightarrow +\infty$. Put

$$a = \log \varphi, \quad Y = \sqrt{5} y, \quad t = Y^{-2}, \quad \theta = \frac{\pi}{a} \log Y, \quad \rho = \frac{\pi}{2a}.$$

Then, for all sufficiently large y ,

$$\Psi(y) = \frac{\log Y}{a} + \frac{1}{2a} \sum_{n \geq 1} t^n Z_n(\theta),$$

where Z_n is a real trigonometric polynomial with Fourier modes $-n, -n+2, \dots, n$. More strongly, with $\lambda = 1 - i\rho$, $X = t^\lambda = te^{i\theta}$, and $A_{jk} = j\lambda + k\bar{\lambda}$, one has the convergent double transseries

$$\Psi(y) = \frac{\log Y}{a} + \frac{1}{2a} \sum_{j+k \geq 1} \frac{(-1)^{j+k+1}}{A_{jk}} \binom{A_{jk}}{j} \binom{A_{jk}}{k} X^j \bar{X}^k.$$

The first six blocks are displayed explicitly, and triangular, fixed-point, coefficient-extraction, and Newton algorithms are developed. A residual certificate converts every truncation into a rigorous numerical approximation.

The Fibonacci example is a specialization of a general inversion theorem. If

$$Y = Ce^{\sigma x} H(c_1 e^{-\alpha_1 x}, \dots, c_r e^{-\alpha_r x}), \quad H(0) = 1,$$

with H analytic and $\operatorname{Re} \alpha_j > 0$, then the inverse correction has the explicit multivariate Lagrange–Good expansion

$$x(Y) = \frac{1}{\sigma} \log \frac{Y}{C} + \frac{1}{\sigma} \sum_{\mathbf{n} \neq 0} \left[-\frac{1}{A_{\mathbf{n}}} [\mathbf{u}^{\mathbf{n}}] H(\mathbf{u})^{A_{\mathbf{n}}} \right] \prod_{j=1}^r \left(c_j (Y/C)^{-\alpha_j/\sigma} \right)^{n_j}, \quad A_{\mathbf{n}} = \sum_j n_j \frac{\alpha_j}{\sigma}.$$

When a combined exponent $A_{\mathbf{n}}$ vanishes, the displayed quotient has a removable singularity: its regularized value is $-\mathbf{u}^{\mathbf{n}} \log H$. Thus the common-factor inversion itself has no genuine resonance pole. The formula unifies real power corrections, logarithmically periodic corrections from complex-conjugate exponents, and quasiperiodic corrections from several incommensurable frequencies. Product and radical perturbations reduce to closed products of generalized binomial coefficients.

Keywords. Fibonacci numbers; Binet continuation; inverse function; transseries; log-periodic asymptotics; complex powers; Lagrange–Good inversion; multivariate Lagrange inversion; exponential–oscillatory asymptotics; series reversion.

Contents

1	The inverse problem and the answer in compact form	4
2	The real Binet–Fibonacci continuation	5
2.1	Equivalent forms	5
2.2	The outer real inverse branches	6
2.3	First-order inversion	7
3	Exact normalization of the inverse equation	7
3.1	The scalar implicit equation	7
3.2	Factorization into two conjugate small monomials	8
3.3	An exact scalar fixed-point identity	9
4	The log-periodic transseries scale	9
4.1	Complex powers and periodic coefficient blocks	9
4.2	Asymptotic meaning of truncation	10
5	Closed coefficients from Lagrange–Good inversion	10
5.1	The coefficient theorem	10
5.2	A compact direct derivation of the first terms	12
6	Explicit coefficient blocks and their structure	12
6.1	The first six blocks	12
6.2	Fourier support, parity, and polynomial dependence	13
6.3	The leading oscillation and multiplicative self-similarity	14
7	Four complementary algorithms	14
7.1	Direct coefficient extraction	14
7.2	Triangular block recurrence	14
7.3	Fixed-point iteration	15
7.4	Newton correction and residual certification	15
8	Analytic convergence and uniform remainders	16
9	Numerical validation	17
10	A universal inversion theorem for exponential analytic perturbations	18
10.1	Set-up and exact reduction	18
10.2	Product and radical perturbations	20
10.3	A conjugate-pair theorem for damped oscillations	21
10.4	Recovery of the Fibonacci formula	21
10.5	Elementary one-variable checks	22
11	The associated transseries algebra	22
11.1	The exponent semigroup	22
11.2	Differentiation and complex powers	23
11.3	Formal inversion theorem	24
11.4	Zero combined exponents and failure of the chosen scale	24

12 Extensions of the method	25
12.1 Several damped harmonics	25
12.2 A polynomial or logarithmic envelope	25
12.3 Lagrange–Bürmann over a coefficient algebra	25
12.4 Complex branches	26
12.5 Other linear recurrences	26
13 Further consequences for the Fibonacci inverse	26
13.1 Sharp first-order oscillation	26
13.2 Ordinary series along geometric phase rays	27
13.3 A phase-averaged elliptic generating function	27
13.4 All logarithmic derivatives	28
13.5 Identities on the Fibonacci lattice	28
14 Computer algebra and numerical implementation	28
14.1 Wolfram Language: triangular block generator	28
14.2 Wolfram Language: inverse approximation and residual	29
14.3 Python/mpmath numerical checker	30
14.4 Implementation cautions	31
15 Research directions and open problems	31
15.1 The exact convergence boundary	32
15.2 Large-order asymptotics of the Fourier blocks	32
15.3 Uniform approximations near the convergence boundary	32
15.4 All real and complex inverse branches	33
15.5 Other recurrences and exponential sums	33
15.6 Formalization	33
15.7 Interaction with broader transseries scales	33
16 Conclusion	33
A Local inverse branches at finite critical values	34
B Coefficient table in complex form	35
C Notation summary	35
D A compact derivation checklist	35

1 The inverse problem and the answer in compact form

The familiar Binet formula interpolates the Fibonacci sequence only after one chooses what to do with the factor $(-1)^x$ away from the integers. One natural complex continuation traces a curve in the complex plane; taking its modulus produces the real, nonnegative function considered here. The oscillation that remains after taking the modulus is exponentially small in x , but it is not negligible for a complete inverse expansion. Upon inversion, it becomes a hierarchy of powers of y^{-2} whose coefficients are periodic functions of $\log y$.

This is precisely the setting in which an ordinary Poincaré series with constant coefficients is too coarse. The correct scale is a *log-periodic transseries*: algebraic decay in the large variable, multiplied by Fourier modes in its logarithm. Equivalently, it is a series in complex powers. The two descriptions are identical because

$$y^{-\alpha-i\beta} = y^{-\alpha} e^{-i\beta \log y}.$$

The directory that motivated this note treats series and transseries as an operational calculus of algebra, composition, and reversion [7]. The present example adds a different scale to that program: a power scale with logarithmically oscillatory coefficient blocks.

Outer inverse and closed all-orders expansion

Set

$$a = \operatorname{arcsch} 2 = \operatorname{arsinh} \frac{1}{2} = \log \varphi, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \rho = \frac{\pi}{2a}. \quad (1)$$

The restriction $\mathcal{F} : [2, \infty) \rightarrow [1, \infty)$ is a bijection. Denote its inverse by Ψ . For $y \rightarrow +\infty$, define

$$Y = \sqrt{5}y, \quad t = Y^{-2}, \quad \theta = \frac{\pi}{a} \log Y, \quad \lambda = 1 - i\rho, \quad X = t^\lambda = te^{i\theta}. \quad (2)$$

For $j, k \geq 0, j + k \geq 1$, put

$$A_{jk} = j\lambda + k\bar{\lambda} = j + k - i\rho(j - k), \quad d_{jk} = \frac{(-1)^{j+k+1}}{A_{jk}} \binom{A_{jk}}{j} \binom{A_{jk}}{k}. \quad (3)$$

Then

$$\Psi(y) = \frac{\log Y}{a} + \frac{1}{2a} \sum_{\substack{j,k \geq 0 \\ j+k \geq 1}} d_{jk} X^j \bar{X}^k. \quad (4)$$

The double series is absolutely convergent for all sufficiently large positive y . Grouping by total degree gives

$$\Psi(y) = \frac{\log Y}{a} + \frac{1}{2a} \sum_{n \geq 1} t^n Z_n(\theta), \quad (5)$$

where

$$Z_n(\theta) = \sum_{j=0}^n \frac{(-1)^{n+1}}{n - i\rho(2j - n)} \binom{n - i\rho(2j - n)}{j} \binom{n - i\rho(2j - n)}{n - j} e^{i(2j-n)\theta}. \quad (6)$$

The negative outer inverse is $\Psi_-(y) = -\Psi(y)$.

The first three nonzero blocks give the immediately usable expansion

$$\Psi(y) = \frac{\log(\sqrt{5}y)}{a} + \frac{\cos \theta}{5ay^2} - \frac{2 + \cos(2\theta) + 2\rho \sin(2\theta)}{50ay^4}$$

$$+ \frac{1}{250ay^6} \left[(6 - \rho^2) \cos \theta + 5\rho \sin \theta + \left(\frac{2}{3} - 3\rho^2 \right) \cos(3\theta) + 3\rho \sin(3\theta) \right] + O(y^{-8}), \quad (7)$$

where $\theta = (\pi/a) \log(\sqrt{5}y)$. The oscillation has constant period $2a$ in $\log Y$, or multiplicative period $e^{2a} = \varphi^2$ in Y , at the level of a fixed Fourier mode. Different blocks carry harmonics of that base phase.

Why a constant-coefficient inverse series cannot be complete

The leading approximation $\Psi(y) \sim a^{-1} \log(\sqrt{5}y)$ is correct, but the first correction is

$$\frac{\cos((\pi/a) \log(\sqrt{5}y))}{5ay^2}.$$

Its coefficient never approaches a constant. Consequently there is no complete expansion of the form $a^{-1} \log(\sqrt{5}y) + \sum_{n \geq 1} c_n y^{-2n}$ with constant c_n . One must retain either periodic coefficient functions or complex powers.

The remainder of the article derives these formulas, proves convergence and branch properties, gives several algorithms, and then develops the general inversion theorem from which (4) follows as a two-factor special case.

2 The real Binet–Fibonacci continuation

2.1 Equivalent forms

We begin by recording the algebraic identities that make the inversion tractable. Throughout,

$$a = \operatorname{arccsch} 2 = \operatorname{arsinh} \frac{1}{2} = \log \varphi > 0. \quad (8)$$

The equality $\operatorname{arsinh}(1/2) = \log \varphi$ follows from

$$\operatorname{arsinh} \frac{1}{2} = \log \left(\frac{1}{2} + \sqrt{1 + \frac{1}{4}} \right) = \log \frac{1 + \sqrt{5}}{2}.$$

Squaring the defining formula gives

$$\begin{aligned} 5\mathcal{F}(x)^2 &= 2 \cosh(2ax) - 2 \cos(\pi x) \\ &= e^{2ax} + e^{-2ax} - 2 \cos(\pi x) \end{aligned} \quad (9)$$

$$= |e^{ax} - e^{-ax} e^{i\pi x}|^2. \quad (10)$$

Thus one may equivalently write

$$\mathcal{F}(x) = \frac{1}{\sqrt{5}} |\varphi^x - \varphi^{-x} e^{i\pi x}|. \quad (11)$$

This is the modulus of the complex Binet continuation studied, in a broader geometric setting, in [6].

Proposition 2.1 (Interpolation, symmetry, and zeros). *The function $\mathcal{F}$ has the following properties.*

1. $\mathcal{F}$ is continuous, nonnegative, and even on $\mathbb{R}$.
2. $\mathcal{F}(x) = 0$ if and only if $x = 0$.
3. For every integer n ,

$$\mathcal{F}(n) = |F_n|, \quad (12)$$

where F_n is the usual two-sided Fibonacci sequence. In particular, $\mathcal{F}(n) = F_n$ for $n \geq 0$.

4. As $x \rightarrow +\infty$,

$$\mathcal{F}(x) = \frac{e^{ax}}{\sqrt{5}} (1 + O(e^{-2ax})). \quad (13)$$

Proof. Evenness and continuity are immediate from the original formula. The modulus representation (10) can vanish only when $e^{ax} = e^{-ax}e^{i\pi x}$. Equality of moduli forces $e^{ax} = e^{-ax}$, hence $x = 0$; conversely $\mathcal{F}(0) = 0$.

At an integer n , $e^{i\pi n} = (-1)^n$, and Binet's formula gives

$$F_n = \frac{\varphi^n - (-1)^n \varphi^{-n}}{\sqrt{5}}.$$

Equation (11) therefore yields $\mathcal{F}(n) = |F_n|$. Finally, factor e^{ax} from (11). The remaining factor is the modulus of $1 - e^{-2ax+i\pi x}$, which equals $1 + O(e^{-2ax})$. $\square$

The modulus in (11) is essential. Without it, the complex continuation retains an exponentially decaying imaginary part. With it, the function is real but its square retains the oscillatory term $-2\cos(\pi x)$. The inverse therefore remembers the phase πx , even though that phase appears only in an exponentially small relative correction to the dominant envelope.

2.2 The outer real inverse branches

The function is not globally one-to-one on $[0, \infty)$; it has small oscillations near the origin. These oscillations are irrelevant for large values, but the branch convention must be stated explicitly.

Lemma 2.2 (A useful exact hyperbolic value). *With $a = \log \varphi$,*

$$\sinh(2a) = \frac{\sqrt{5}}{2}, \quad \cosh(2a) = \frac{3}{2}, \quad \sinh(4a) = \frac{3\sqrt{5}}{2}. \quad (14)$$

Moreover,

$$3\sqrt{5}a > \pi. \quad (15)$$

Proof. Since $\varphi^2 = (3 + \sqrt{5})/2$ and $\varphi^{-2} = (3 - \sqrt{5})/2$, the first two identities follow by taking their half-difference and half-sum. The double-angle identity then gives $\sinh(4a) = 2\sinh(2a)\cosh(2a) = 3\sqrt{5}/2$.

For a completely elementary proof of the inequality, use $(1 + s^2)^{-1/2} \geq 1 - s^2/2$ for $0 \leq s \leq 1/2$. Hence

$$a = \int_0^{1/2} \frac{ds}{\sqrt{1+s^2}} \geq \frac{1}{2} - \frac{1}{48} = \frac{23}{48}.$$

Therefore $3\sqrt{5}a \geq 23\sqrt{5}/16$. The classical bound $\pi < 22/7$, together with $23\sqrt{5}/16 > 22/7$, proves (15). The latter comparison may be checked by squaring: $5 \cdot 161^2 > 352^2$. $\square$

Theorem 2.3 (Monotonicity of the outer branch). *The restriction*

$$\mathcal{F} : [2, \infty) \longrightarrow [1, \infty) \quad (16)$$

is a strictly increasing bijection. Consequently there is a unique outer positive inverse

$$\Psi : [1, \infty) \longrightarrow [2, \infty), \quad \mathcal{F}(\Psi(y)) = y. \quad (17)$$

By evenness, $\Psi_-(y) = -\Psi(y)$ is the outer negative inverse.

Proof. Let $G(x) = \cosh(2ax) - \cos(\pi x)$. Since $\mathcal{F}$ is a positive constant times $\sqrt{G}$, its derivative has the sign of

$$G'(x) = 2a \sinh(2ax) + \pi \sin(\pi x).$$

For $x \geq 2$, [Theorem 2.2](#) gives

$$2a \sinh(2ax) \geq 2a \sinh(4a) = 3\sqrt{5}a > \pi,$$

whereas $\pi \sin(\pi x) \geq -\pi$. Thus $G'(x) > 0$ and hence $\mathcal{F}'(x) > 0$ on $[2, \infty)$. Also $\mathcal{F}(2) = F_2 = 1$, while $\mathcal{F}(x) \rightarrow \infty$ by (13). The asserted bijection follows. Evenness supplies the negative branch. $\square$

Remark 2.4 (How many real inverse values?). The compact interval $[-2, 2]$ can contribute additional inverse values only for bounded y . A simple rigorous bound is

$$\max_{|x| \leq 2} \mathcal{F}(x) \leq \sqrt{\frac{2}{5}(\cosh(4a) + 1)} = \frac{3}{\sqrt{5}}.$$

Hence, for $y > 3/\sqrt{5}$, the only real solutions of $\mathcal{F}(x) = y$ are $x = \pm\Psi(y)$. Numerically, the actual final internal local maximum is much smaller: it occurs near

$$x = 1.1323790680, \quad \mathcal{F}(x) = 1.0138598855,$$

and is followed by a local minimum near

$$x = 1.7192508860, \quad \mathcal{F}(x) = 0.9111726845.$$

These decimal values are descriptive, not needed in any proof below.

2.3 First-order inversion

The envelope (13) immediately suggests

$$\Psi(y) \sim \frac{1}{a} \log(\sqrt{5}y). \quad (18)$$

The central problem is to calculate the complete correction. The relative perturbation in $\mathcal{F}(x)$ has size e^{-2ax} ; after substituting the leading inverse, this becomes $(\sqrt{5}y)^{-2}$. The phase πx becomes $(\pi/a) \log(\sqrt{5}y)$. This dimensional argument already predicts a series in y^{-2} with periodic coefficients in $\log y$. The next section turns this prediction into an exact normalized equation.

3 Exact normalization of the inverse equation

3.1 The scalar implicit equation

Let $y \geq 1$ and set $x = \Psi(y)$. Introduce

$$Y = \sqrt{5}y, \quad x_0 = \frac{\log Y}{a}, \quad t = Y^{-2}, \quad \theta = \frac{\pi}{a} \log Y, \quad \rho = \frac{\pi}{2a}. \quad (19)$$

Write the inverse in the anchored form

$$x = x_0 + \frac{z}{2a}. \quad (20)$$

The unknown $z = z(t, \theta)$ is small when t is small.

Proposition 3.1 (Exact normalized equation). *The correction z satisfies*

$$\boxed{H(z; t, \theta) := e^z + t^2 e^{-z} - 2t \cos(\theta + \rho z) - 1 = 0.} \quad (21)$$

Conversely, the solution of (21) that tends to zero with t yields the outer inverse through (20).

Proof. The equation $\mathcal{F}(x) = y$, after squaring and multiplying by 5, is

$$Y^2 = e^{2ax} + e^{-2ax} - 2 \cos(\pi x).$$

Under (20),

$$e^{2ax} = Y^2 e^z, \quad e^{-2ax} = Y^{-2} e^{-z}, \quad \pi x = \theta + \rho z.$$

Division by Y^2 , with $t = Y^{-2}$, gives (21). At $(z, t) = (0, 0)$, one has $H = 0$ and $H_z = 1$, so the analytic implicit-function theorem selects a unique small solution for t near zero. This solution corresponds to $x \rightarrow +\infty$ and hence to the outer branch. $\square$

The exact derivative needed for Newton iteration and error control is

$$H_z(z; t, \theta) = e^z - t^2 e^{-z} + 2\rho t \sin(\theta + \rho z). \quad (22)$$

The key point is that $H_z \rightarrow 1$ uniformly in the real phase θ as $t \rightarrow 0$. Thus the implicit solution exists not merely for each fixed phase, but uniformly over the phase circle when t is sufficiently small.

3.2 Factorization into two conjugate small monomials

The scalar equation is best solved after exposing its two complex-conjugate factors. From (9),

$$Y^2 = e^{2ax} (1 - e^{(-2a+i\pi)x}) (1 - e^{(-2a-i\pi)x}). \quad (23)$$

Define

$$q = e^{-2ax} = t e^{-z}, \quad \lambda = 1 - i\rho, \quad \bar{\lambda} = 1 + i\rho. \quad (24)$$

Because $2a\rho = \pi$, the oscillatory factors are

$$u = q^\lambda = e^{(-2a+i\pi)x}, \quad v = q^{\bar{\lambda}} = e^{(-2a-i\pi)x}. \quad (25)$$

On the positive real branch, $v = \bar{u}$. Temporarily treating them as independent analytic variables is the decisive simplification.

Equation (23) is equivalent to

$$q = t(1 - u)(1 - v). \quad (26)$$

Let

$$X = t^\lambda = t e^{i\theta}, \quad V = t^{\bar{\lambda}} = t e^{-i\theta}. \quad (27)$$

Raising (26) to the powers λ and $\bar{\lambda}$ gives the coupled system

$$\boxed{\begin{aligned} u &= X(1 - u)^\lambda (1 - v)^\lambda, \\ v &= V(1 - u)^{\bar{\lambda}} (1 - v)^{\bar{\lambda}}. \end{aligned}} \quad (28)$$

Moreover, since $q/t = e^{-z}$,

$$\boxed{z = -\log((1 - u)(1 - v)).} \quad (29)$$

Equations (28)–(29) are the exact Lagrange–Good form of the inverse problem.

The normalization pattern

The example illustrates a reusable sequence of steps.

1. Separate the dominant exponential envelope and invert it exactly.
2. Measure the inverse displacement by a dimensionless variable z .
3. Factor every trigonometric perturbation into conjugate exponentials.
4. Promote the conjugate small monomials to independent variables.
5. Apply multivariate Lagrange inversion; impose conjugacy only at the end.

This converts a nonlinear phase shift such as $\cos(\theta + \rho z)$ into an algebraic multivariate fixed-point system.

3.3 An exact scalar fixed-point identity

Expanding the logarithms in (29), while retaining $q = te^{-z}$, gives another useful exact representation:

$$\begin{aligned} z &= \sum_{m \geq 1} \frac{u^m + v^m}{m} \\ &= 2 \sum_{m \geq 1} \frac{t^m}{m} e^{-mz} \cos(m\theta + m\rho z). \end{aligned} \quad (30)$$

For the physical positive branch, $0 < q < 1$, so the logarithmic series is absolutely convergent. Equation (30) is convenient for numerical fixed-point iteration and for intuitive power counting. The multivariate form (28), however, is what produces the closed coefficient formula.

4 The log-periodic transseries scale

4.1 Complex powers and periodic coefficient blocks

For nonnegative integers j, k ,

$$X^j V^k = t^{j\lambda + k\bar{\lambda}} = t^{j+k} e^{i(j-k)\theta}. \quad (31)$$

Thus total degree $n = j + k$ fixes the algebraic size $t^n = Y^{-2n}$, while $j - k$ determines the Fourier harmonic. This leads to the following terminology.

Definition 4.1 (Log-periodic power transseries). A log-periodic power transseries in $t \rightarrow 0^+$ is an expression

$$S(t) = \sum_{n \geq n_0} t^n P_n(\log t), \quad (32)$$

where each P_n is a finite trigonometric polynomial in $\log t$. It may equivalently be written as a locally finite series of complex powers $\sum c_\alpha t^\alpha$, with exponents occurring in conjugate pairs.

In the present normalization, $\theta = -(\pi/(2a)) \log t = -\rho \log t$. Hence a Fourier factor $e^{im\theta}$ is the complex power $t^{-im\rho}$. The natural monomial semigroup is

$$\mathcal{M} = \left\{ t^{j\lambda + k\bar{\lambda}} : j, k \in \mathbb{N} \right\}. \quad (33)$$

Every nonconstant monomial has positive real exponent $j + k$, so only finitely many monomials lie above any prescribed asymptotic size. This local finiteness is what makes multiplication, composition with analytic germs, and coefficientwise reversion well defined.

Proposition 4.2 (Block structure). *Suppose*

$$z(X, V) = \sum_{j+k \geq 1} d_{jk} X^j V^k.$$

On the physical locus $V = \overline{X}$, it has the block decomposition

$$z(t, \theta) = \sum_{n \geq 1} t^n Z_n(\theta), \quad Z_n(\theta) = \sum_{j=0}^n d_{j, n-j} e^{i(2j-n)\theta}. \quad (34)$$

If $d_{kj} = \overline{d_{jk}}$, then every Z_n is real-valued. Its Fourier support is

$$\{-n, -n+2, \dots, n-2, n\}. \quad (35)$$

Consequently

$$Z_n(\theta + \pi) = (-1)^n Z_n(\theta). \quad (36)$$

Proof. Substitute (31) and group terms with $j+k=n$. Complex conjugation interchanges j and k , proving reality under the stated symmetry. Since $2j-n$ runs from $-n$ to n in steps of two, the support and parity follow. $\square$

4.2 Asymptotic meaning of truncation

Because $|X| = |V| = t$, truncating by total degree is the same as truncating by asymptotic order. If the bivariate series is analytic in a polydisc $|X|, |V| < R$, then for every fixed $r < R$, Cauchy estimates give a constant C_r such that

$$\left| z(t, \theta) - \sum_{n=1}^N t^n Z_n(\theta) \right| \leq C_r t^{N+1} \quad (37)$$

for $0 \leq t \leq r' < r$, uniformly in θ . After dividing by $2a$, the inverse remainder is $O(Y^{-2N-2}) = O(y^{-2N-2})$.

This is stronger than a purely formal asymptotic statement: the transseries will be shown to converge for sufficiently large y . Nevertheless the formal viewpoint remains useful because the same coefficient formulas survive in wider settings where convergence may be unavailable or where additional asymptotic scales are present.

5 Closed coefficients from Lagrange–Good inversion

5.1 The coefficient theorem

For $w \in \mathbb{C}$ and $m \in \mathbb{N}$, use the generalized binomial coefficient

$$\binom{w}{m} = \frac{w(w-1) \cdots (w-m+1)}{m!}, \quad \binom{w}{0} = 1. \quad (38)$$

The multivariate inversion formula used below goes back to Good [5]; see also [3, 4, 8] for modern proofs and variants.

Theorem 5.1 (Closed coefficient formula). *Let (u, v) be the analytic solution near $(X, V) = (0, 0)$ of (28), and let z be given by (29). Then*

$$z(X, V) = \sum_{\substack{j, k \geq 0 \\ j+k \geq 1}} d_{jk} X^j V^k, \quad (39)$$

where

$$d_{jk} = \frac{(-1)^{j+k+1}}{A_{jk}} \binom{A_{jk}}{j} \binom{A_{jk}}{k}, \quad A_{jk} = j\lambda + k\bar{\lambda}. \quad (40)$$

The series is convergent in a neighborhood of $(X, V) = (0, 0)$.

Proof. The right-hand sides of (28) are analytic near the origin when the principal germs of the powers are chosen. The Jacobian of the system with respect to (u, v) is the identity at the origin, so the analytic implicit-function theorem produces a unique analytic solution with $u(0, 0) = v(0, 0) = 0$.

Write

$$\phi_1(u, v) = (1 - u)^\lambda (1 - v)^\lambda, \quad \phi_2(u, v) = (1 - u)^{\bar{\lambda}} (1 - v)^{\bar{\lambda}}. \quad (41)$$

The two-variable Lagrange–Good formula says that, for any analytic germ $K(u, v)$,

$$[X^j V^k] K(u(X, V), v(X, V)) = [u^j v^k] K(u, v) \phi_1(u, v)^j \phi_2(u, v)^k \Delta(u, v), \quad (42)$$

where

$$\Delta = \det \left(\delta_{rs} - w_s \frac{\partial}{\partial w_s} \log \phi_r \right)_{r,s=1}^2, \quad (w_1, w_2) = (u, v). \quad (43)$$

For $K = z = -\log((1 - u)(1 - v))$, one has

$$z_u = \frac{1}{1 - u}, \quad z_v = \frac{1}{1 - v}. \quad (44)$$

Since $\log \phi_1 = -\lambda z$ and $\log \phi_2 = -\bar{\lambda} z$, the determinant is rank one and reduces to

$$\Delta = 1 + \lambda u z_u + \bar{\lambda} v z_v. \quad (45)$$

For fixed $(j, k) \neq (0, 0)$, put

$$A = A_{jk} = j\lambda + k\bar{\lambda}, \quad P = \phi_1^j \phi_2^k = (1 - u)^A (1 - v)^A = e^{-Az}. \quad (46)$$

The following elementary derivative identity is the cancellation at the heart of the computation:

$$u \frac{\partial}{\partial u} \left[\left(z + \frac{1}{A} \right) P \right] = -A z u z_u P, \quad (47)$$

and the analogous identity holds with u replaced by v . Extracting $[u^j v^k]$ and using $[u^j v^k] u \partial_u Q = j [u^j v^k] Q$, we obtain

$$\begin{aligned} [u^j v^k] z P \Delta &= [u^j v^k] z P - \frac{\lambda j + \bar{\lambda} k}{A} [u^j v^k] \left(z + \frac{1}{A} \right) P \\ &= -\frac{1}{A} [u^j v^k] P. \end{aligned} \quad (48)$$

Finally,

$$[u^j v^k] P = (-1)^{j+k} \binom{A}{j} \binom{A}{k}. \quad (49)$$

Substitution into (42) gives (40). $\square$

Remark 5.2 (Why the Fibonacci denominator never vanishes). For $(j, k) \neq (0, 0)$,

$$\operatorname{Re} A_{jk} = j + k > 0.$$

Hence $A_{jk} \neq 0$. The universal theorem in [Section 10](#) also covers zero combined exponents by a removable continuation, but the Fibonacci specialization never needs that continuation.

Corollary 5.3 (The outer inverse transseries). *On the physical locus $V = \overline{X}$, the coefficients satisfy*

$$d_{kj} = \overline{d_{jk}}, \quad (50)$$

and (39) is real. Substitution into (20) yields (4), while grouping by $n = j + k$ yields (5)–(6).

Proof. Conjugation interchanges λ and $\overline{\lambda}$, hence interchanges A_{jk} and A_{kj} ; generalized binomial coefficients respect conjugation. This proves (50). The remaining statements follow from (31). $\square$

5.2 A compact direct derivation of the first terms

Although the closed formula is the cleanest all-orders result, it is useful to see the first correction arise directly from (21). Suppose

$$z = tZ_1(\theta) + t^2Z_2(\theta) + O(t^3).$$

At order t ,

$$Z_1 - 2\cos\theta = 0,$$

so $Z_1 = 2\cos\theta$. At order t^2 , expansion of the exponential and of the shifted cosine gives

$$Z_2 + \frac{1}{2}Z_1^2 + 1 + 2\rho Z_1 \sin\theta = 0.$$

Using $Z_1 = 2\cos\theta$ and elementary double-angle identities yields

$$Z_2 = -2 - \cos(2\theta) - 2\rho \sin(2\theta).$$

This computation already shows how nonlinear phase modulation creates sine terms even though the original defining equation contains only a cosine.

6 Explicit coefficient blocks and their structure

6.1 The first six blocks

The all-orders formula (6) can be expanded into real trigonometric polynomials. The first six are as follows:

$$Z_1(\theta) = 2\cos\theta, \quad (51)$$

$$Z_2(\theta) = -2 - \cos(2\theta) - 2\rho \sin(2\theta), \quad (52)$$

$$\begin{aligned} Z_3(\theta) &= (6 - \rho^2)\cos\theta + 5\rho \sin\theta \\ &\quad + \left(\frac{2}{3} - 3\rho^2\right)\cos(3\theta) + 3\rho \sin(3\theta), \end{aligned} \quad (53)$$

$$\begin{aligned} Z_4(\theta) &= -9 + 4(3\rho^2 - 2)\cos(2\theta) + \frac{4\rho(2\rho^2 - 13)}{3}\sin(2\theta) \\ &\quad + \frac{16\rho^2 - 1}{2}\cos(4\theta) + \frac{\rho(16\rho^2 - 11)}{3}\sin(4\theta), \end{aligned} \quad (54)$$

$$\begin{aligned} Z_5(\theta) &= \frac{\rho^4 - 95\rho^2 + 240}{6}\cos\theta - \frac{4\rho(2\rho^2 - 31)}{3}\sin\theta \\ &\quad + \frac{27\rho^4 - 213\rho^2 + 40}{4}\cos(3\theta) - \frac{7\rho(9\rho^2 - 11)}{2}\sin(3\theta) \\ &\quad + \frac{625\rho^4 - 875\rho^2 + 24}{60}\cos(5\theta) - \frac{25\rho(5\rho^2 - 1)}{6}\sin(5\theta), \end{aligned} \quad (55)$$

$$\begin{aligned}
Z_6(\theta) = & -\frac{200}{3} - \frac{92\rho^4 - 937\rho^2 + 450}{6} \cos(2\theta) \\
& - \frac{\rho(8\rho^4 - 418\rho^2 + 1035)}{6} \sin(2\theta) - \frac{4(64\rho^4 - 116\rho^2 + 9)}{3} \cos(4\theta) \\
& - \frac{4\rho(64\rho^4 - 620\rho^2 + 261)}{15} \sin(4\theta) - \frac{324\rho^4 - 135\rho^2 + 2}{6} \cos(6\theta) \\
& - \frac{\rho(648\rho^4 - 1530\rho^2 + 137)}{30} \sin(6\theta).
\end{aligned} \tag{56}$$

Numerically,

$$a = 0.481211825059603 \dots, \quad \rho = 3.26425130263650 \dots \tag{57}$$

The coefficients can therefore be sizable even though the accompanying factor $t^n = (5y^2)^{-n}$ decays rapidly.

Six-block approximation

Define

$$\Psi^{[N]}(y) = \frac{\log(\sqrt{5}y)}{a} + \frac{1}{2a} \sum_{n=1}^N \frac{Z_n(\theta)}{(5y^2)^n}, \quad \theta = \frac{\pi}{a} \log(\sqrt{5}y).$$

Then $\Psi(y) - \Psi^{[N]}(y) = O(y^{-2N-2})$ as $y \rightarrow \infty$, uniformly with respect to the phase generated by y . Equations (51)–(56) provide a direct approximation through order y^{-12} , with remainder $O(y^{-14})$.

6.2 Fourier support, parity, and polynomial dependence

The closed coefficient formula makes several structural laws immediate.

Proposition 6.1 (Structural laws for Z_n). *For every $n \geq 1$:*

1. Z_n is a real trigonometric polynomial with the Fourier support (35).
2. $Z_n(\theta + \pi) = (-1)^n Z_n(\theta)$.
3. Written in the real basis $1, \cos(m\theta), \sin(m\theta)$, every coefficient is a polynomial in ρ of degree at most $n - 1$, with rational coefficients.
4. A constant Fourier term occurs only when $n = 2m$ is even, and then

$$\langle Z_{2m} \rangle := \frac{1}{2\pi} \int_0^{2\pi} Z_{2m}(\theta) \, d\theta = -\frac{1}{2m} \binom{2m}{m}. \tag{58}$$

For odd n , $\langle Z_n \rangle = 0$.

Proof. The first two statements were proved abstractly in Theorem 4.2. Since $A_{jk} = n - i\rho(2j - n)$, each d_{jk} is a polynomial in A_{jk} of degree at most $j + k - 1 = n - 1$: the apparent denominator is canceled by the factor A_{jk} contained in at least one of the two binomial coefficients. Pairing conjugate terms produces rational polynomials in ρ .

The zero Fourier mode requires $2j - n = 0$, so $n = 2m$ and $j = k = m$. In that case $A_{mm} = 2m$, and (40) gives

$$d_{mm} = -\frac{1}{2m} \binom{2m}{m}^2.$$

No other term contributes to the phase average. □

The mean correction has an interesting combinatorial size. The central binomial estimate $\binom{2m}{m} \sim 4^m / \sqrt{\pi m}$ gives

$$\langle Z_{2m} \rangle \sim -\frac{16^m}{2\pi m^2}. \quad (59)$$

Thus the even, phase-averaged subseries in t has radius at most $1/4$. This does not determine the exact convergence boundary of the full physical series, but it gives a sharp structural warning: the convergent expansion is a large-argument expansion, not a globally convergent representation down to $y = 1$.

6.3 The leading oscillation and multiplicative self-similarity

The first correction is

$$\Delta_1(y) = \frac{1}{a} t \cos \theta = \frac{1}{5ay^2} \cos\left(\frac{\pi}{a} \log(\sqrt{5}y)\right). \quad (60)$$

Multiplication of Y by $\varphi^2 = e^{2a}$ advances θ by 2π , while it multiplies t by φ^{-4} . Hence the shape of the leading oscillation repeats on a geometric progression, with a fixed amplitude scaling. More generally, the block $t^n Z_n(\theta)$ transforms by

$$t^n Z_n(\theta) \mapsto \varphi^{-4n} t^n Z_n(\theta) \quad \text{under} \quad Y \mapsto \varphi^2 Y. \quad (61)$$

This is a discrete scale-covariance law, a common signature of complex asymptotic exponents.

7 Four complementary algorithms

The closed formula is ideal for theory, but different tasks favor different computational forms. This section records four mutually checking algorithms.

7.1 Direct coefficient extraction

Algorithm 7.1 (Closed-coefficient construction). To construct the expansion through total degree N :

1. Set $\lambda = 1 - i\rho$.
2. For every $n = 1, \dots, N$ and $j = 0, \dots, n$, put

$$k = n - j, \quad A_{jk} = j\lambda + k\bar{\lambda}, \quad d_{jk} = \frac{(-1)^{n+1}}{A_{jk}} \binom{A_{jk}}{j} \binom{A_{jk}}{k}.$$

3. Form

$$Z_n(\theta) = \sum_{j=0}^n d_{j,n-j} e^{i(2j-n)\theta}.$$

4. Pair the j and $n - j$ terms to convert the answer to a manifestly real sine-cosine form.

Only $(N+1)(N+2)/2 - 1$ coefficient pairs are needed. For symbolic work, the integer lower arguments of the binomial coefficients make each coefficient a finite polynomial computation; no infinite expansion is hidden in (40).

7.2 Triangular block recurrence

The normalized scalar equation produces a particularly simple recurrence. Suppose $Z_1, \dots, Z_{n-1}$ are known and define

$$z_{<n}(t, \theta) = \sum_{r=1}^{n-1} t^r Z_r(\theta). \quad (62)$$

Proposition 7.2 (Triangular recurrence). *For $n \geq 1$,*

$$Z_n(\theta) = -[t^n] \{ e^{z_{<n}} + t^2 e^{-z_{<n}} - 2t \cos(\theta + \rho z_{<n}) - 1 \}. \quad (63)$$

Proof. Insert $z = z_{<n} + t^n Z_n + O(t^{n+1})$ into (21). At order t^n , the new term contributes Z_n through e^z . Its contribution through $t^2 e^{-z}$ starts at order t^{n+2} , and through the shifted cosine at order t^{n+1} . Therefore the coefficient of t^n is Z_n plus the coefficient computed from $z_{<n}$, proving the formula. $\square$

The recurrence is well suited to a computer algebra system because it never requires solving a nonlinear equation at a given order. It also proves by induction that each block is a finite trigonometric polynomial.

7.3 Fixed-point iteration

Equation (30) defines

$$\Phi_{t,\theta}(z) = 2 \sum_{m \geq 1} \frac{t^m}{m} e^{-mz} \cos(m\theta + m\rho z). \quad (64)$$

Starting from $z^{(0)} = 0$, one may iterate

$$z^{(r+1)} = \Phi_{t,\theta}(z^{(r)}). \quad (65)$$

For sufficiently small t , this is a contraction on a phase-independent interval. For example, if $|z| \leq R$ and $te^R < 1$, then

$$|\Phi_{t,\theta}(z)| \leq -2 \log(1 - te^R), \quad (66)$$

$$|\Phi'_{t,\theta}(z)| \leq 2\sqrt{1 + \rho^2} \frac{te^R}{1 - te^R}. \quad (67)$$

Thus any pair (t, R) for which the right side of (66) is at most R and the right side of (67) is less than one gives a rigorous contraction domain. This method evaluates the exact correction but does not by itself expose the closed coefficient structure.

7.4 Newton correction and residual certification

Given any approximation z_0 , one Newton step for the exact scalar equation is

$$z_1 = z_0 - \frac{H(z_0; t, \theta)}{H_z(z_0; t, \theta)}, \quad (68)$$

with H_z from (22). A truncated transseries makes an excellent initial value: if $z_0 = \sum_{n=1}^N t^n Z_n$, then $H(z_0; t, \theta) = O(t^{N+1})$, and one Newton step normally improves the formal order to approximately $O(t^{2N+2})$.

Theorem 7.3 (Elementary residual certificate). *Let $z_0 \in \mathbb{R}$, $r > 0$, and suppose*

$$H_z(z; t, \theta) \geq m > 0 \quad \text{for every } z \in [z_0 - r, z_0 + r]. \quad (69)$$

If

$$|H(z_0; t, \theta)| \leq mr, \quad (70)$$

then (21) has a unique root z_ in that interval and*

$$|z_* - z_0| \leq \frac{|H(z_0; t, \theta)|}{m}. \quad (71)$$

Consequently the inverse error is at most $|H(z_0; t, \theta)| / (2am)$.

Proof. By the mean-value theorem,

$$H(z_0 + r) \geq H(z_0) + mr \geq 0, \quad H(z_0 - r) \leq H(z_0) - mr \leq 0.$$

Continuity gives a root, and strict positivity of H_z gives uniqueness. Applying the mean-value theorem between z_0 and the root yields (71). $\square$

This certificate is especially useful in rigorous computation: interval arithmetic can bound H_z on a short interval around the truncation, while the residual is evaluated directly from the original equation. No estimate of all omitted coefficients is required.

8 Analytic convergence and uniform remainders

The word “transseries” often suggests a formal or divergent expansion. In the present problem the situation is better: the relevant transseries is the restriction of a convergent bivariate Taylor series.

Theorem 8.1 (Convergence for large arguments). *There is an $R > 0$ such that the solution $z(X, V)$ of (28)–(29) is analytic for $|X| < R$, $|V| < R$. Consequently, for every sufficiently large positive y , the series (4) and (5) converge absolutely to the outer inverse $\Psi(y)$. The convergence is uniform in the phase θ whenever $0 \leq t \leq r < R$.*

Proof. Analyticity near the origin was established at the start of the proof of Theorem 5.1. Hence there is a polydisc of analyticity $|X| < R$, $|V| < R$, possibly after decreasing R . On the physical locus, $|X| = |V| = t$, independently of θ . Therefore every $t < R$ lies in the polydisc for all real phases, and absolute, locally uniform convergence of the Taylor series follows. The analytic solution tends to zero as $t \rightarrow 0$, so by the branch selection in Theorem 3.1 it is the correction belonging to the outer inverse. $\square$

Proposition 8.2 (A Cauchy-type tail bound). *Choose $R_0 > 0$ such that z is analytic on a neighborhood of the closed polydisc $|X|, |V| \leq R_0$, and put*

$$M_{R_0} = \max_{|X|, |V| \leq R_0} |z(X, V)|.$$

For $0 < t < R_0$, let $q = t/R_0$. Then

$$\begin{aligned} \left| z(t, \theta) - \sum_{n=1}^N t^n Z_n(\theta) \right| &\leq M_{R_0} \sum_{n=N+1}^{\infty} (n+1) q^n \\ &= M_{R_0} \frac{q^{N+1} ((N+2) - (N+1)q)}{(1-q)^2}. \end{aligned} \quad (72)$$

The bound is uniform in θ .

Proof. Cauchy’s estimate in two variables gives $|d_{jk}| \leq M_{R_0} R_0^{-j-k}$. There are $n+1$ pairs (j, k) with $j+k=n$. On the physical locus each corresponding monomial has modulus t^n . Summing the geometric-arithmetical tail proves the result. $\square$

Corollary 8.3 (Poincaré remainder). *For every fixed $N \geq 0$,*

$$\Psi(y) = \frac{\log(\sqrt{5}y)}{a} + \frac{1}{2a} \sum_{n=1}^N \frac{Z_n(\theta)}{(5y^2)^n} + O(y^{-2N-2}), \quad y \rightarrow +\infty. \quad (73)$$

The implied constant can be chosen uniformly over the phase circle.

Remark 8.4 (The maximal convergence domain). The analytic implicit-function theorem proves a nonzero convergence domain but does not locate its nearest singularity. The maximal bivariate domain is bounded by the singular variety on which either the fixed-point solution encounters a branch singularity $u = 1$ or $v = 1$, or the Lagrange–Good Jacobian determinant vanishes. Determining which component first meets the physical curve $(X, V) = (te^{i\theta}, te^{-i\theta})$ is an interesting separate problem; see [Section 15](#).

9 Numerical validation

The formulas can be checked without relying on the expansion itself: solve $\mathcal{F}(x) = y$ numerically on $[2, \infty)$, evaluate the truncated blocks, and compare. Let

$$\Psi^{[N]}(y) = \frac{\log(\sqrt{5}y)}{a} + \frac{1}{2a} \sum_{n=1}^N \frac{Z_n(\theta)}{(5y^2)^n}, \quad (74)$$

with $\Psi^{[0]} = a^{-1} \log(\sqrt{5}y)$. The following table reports absolute errors. The displayed values were computed with high-precision arithmetic from the original defining equation.

Table 1: Absolute errors $|\Psi^{[N]}(y) - \Psi(y)|$.

y	$N = 0$	$N = 1$	$N = 2$	$N = 4$	$N = 6$
2	1.12696×10^{-1}	1.52360×10^{-2}	2.96187×10^{-3}	3.76916×10^{-4}	1.28682×10^{-4}
5	1.68278×10^{-2}	2.26345×10^{-4}	1.85965×10^{-5}	2.98868×10^{-8}	1.82331×10^{-11}
10	5.45185×10^{-4}	1.13522×10^{-5}	1.54252×10^{-7}	3.01709×10^{-11}	6.00018×10^{-15}
50	1.35073×10^{-4}	2.57013×10^{-8}	7.18834×10^{-12}	8.65363×10^{-19}	9.96184×10^{-25}
100	3.01016×10^{-5}	3.56196×10^{-9}	2.97159×10^{-13}	4.30138×10^{-21}	1.48686×10^{-28}

For reference, the corresponding outer inverse values are

$$\begin{aligned} \Psi(2) &= 3, & \Psi(5) &= 5, \\ \Psi(10) &= 6.45779309028821164945\dots, & \Psi(50) &= 9.80193485400697171349\dots, \\ \Psi(100) &= 11.24218977013280389416\dots \end{aligned}$$

The exact equalities $\Psi(2) = 3$ and $\Psi(5) = 5$ are consequences of the integer interpolation. More generally,

$$\boxed{\Psi(F_n) = n \quad (n \geq 2)}. \quad (75)$$

At $y = 2$ the small parameter is only $t = 1/20$, so convergence is slower and individual errors need not decrease by a fixed ratio. Already by $y = 5$, six blocks deliver about eleven correct decimal digits; at larger arguments the algebraic hierarchy becomes extremely effective.

Remark 9.1 (A simple consistency check). The truncation residual

$$R_N(y) = H\left(\sum_{n=1}^N t^n Z_n(\theta); t, \theta\right)$$

must be $O(t^{N+1})$. Expanding symbolically verifies that the coefficients of $t, t^2, \dots, t^N$ vanish identically. Evaluating R_N numerically and applying [Theorem 7.3](#) gives an independent a posteriori check of every row in [Table 1](#).

10 A universal inversion theorem for exponential analytic perturbations

The preceding derivation is not tied to Fibonacci numbers. Its essential features are a dominant exponential envelope and analytic dependence on one or more exponentially small monomials. We now formulate and prove the general result.

10.1 Set-up and exact reduction

Let $r \geq 1$, let $H(\mathbf{u})$ be analytic near $\mathbf{u} = \mathbf{0} \in \mathbb{C}^r$, and assume

$$H(\mathbf{0}) = 1. \quad (76)$$

Choose the analytic germ of $\log H$ that vanishes at the origin. Consider

$$Y = Ce^{\sigma x} H(c_1 e^{-\alpha_1 x}, \dots, c_r e^{-\alpha_r x}), \quad (77)$$

where $C \neq 0$, $\sigma \neq 0$, and branches of all logarithms and powers are fixed. For a real outer inverse one normally assumes

$$C > 0, \quad \sigma > 0, \quad \operatorname{Re} \alpha_j > 0. \quad (78)$$

Define

$$L = \operatorname{Log}(Y/C), \quad \lambda_j = \frac{\alpha_j}{\sigma}, \quad X_j = c_j e^{-\lambda_j L} = c_j (Y/C)^{-\lambda_j}. \quad (79)$$

Write

$$x = \frac{L + z}{\sigma}. \quad (80)$$

Then the small internal variables are

$$u_j = c_j e^{-\alpha_j x} = X_j e^{-\lambda_j z}. \quad (81)$$

Substitution into (77) gives the exact relations

$$z = -\log H(\mathbf{u}), \quad u_j = X_j H(\mathbf{u})^{\lambda_j} \quad (1 \leq j \leq r). \quad (82)$$

This is an r -variable Lagrange–Good system with a common analytic factor. The common-factor structure is what causes the determinant and the final coefficient formula to collapse.

For a multi-index $\mathbf{n} = (n_1, \dots, n_r) \in \mathbb{N}^r$, write

$$|\mathbf{n}| = n_1 + \dots + n_r, \quad \mathbf{X}^{\mathbf{n}} = \prod_{j=1}^r X_j^{n_j}, \quad A_{\mathbf{n}} = \mathbf{n} \cdot \boldsymbol{\lambda} = \sum_{j=1}^r n_j \lambda_j. \quad (83)$$

For any scalar A , define the *regularized power quotient*

$$\mathcal{Q}_A[H](\mathbf{u}) := \sum_{m \geq 1} \frac{A^{m-1}}{m!} (\log H(\mathbf{u}))^m = \begin{cases} \frac{H(\mathbf{u})^A - 1}{A}, & A \neq 0, \\ \log H(\mathbf{u}), & A = 0. \end{cases} \quad (84)$$

Because $\log H$ has zero constant term, only finitely many summands in (84) contribute to any fixed coefficient. In particular, coefficient extraction from $\mathcal{Q}_A[H]$ is an entire, in fact polynomial, function of A .

Theorem 10.1 (Universal coefficient formula). *Assume H is analytic near the origin and $H(\mathbf{0}) = 1$. Then the system (82) has a unique analytic solution near $\mathbf{X} = \mathbf{0}$, and*

$$z(\mathbf{X}) = \sum_{\mathbf{n} \neq \mathbf{0}} b_{\mathbf{n}} \mathbf{X}^{\mathbf{n}}, \quad (85)$$

with the explicit, everywhere regular coefficients

$$b_{\mathbf{n}} = -[\mathbf{u}^{\mathbf{n}}] \mathcal{Q}_{A_{\mathbf{n}}}[H](\mathbf{u}). \quad (86)$$

Equivalently,

$$b_{\mathbf{n}} = \begin{cases} -\frac{1}{A_{\mathbf{n}}} [\mathbf{u}^{\mathbf{n}}] H(\mathbf{u})^{A_{\mathbf{n}}}, & A_{\mathbf{n}} \neq 0, \\ -[\mathbf{u}^{\mathbf{n}}] \log H(\mathbf{u}), & A_{\mathbf{n}} = 0. \end{cases} \quad (87)$$

Consequently the inverse of (77) is

$$x(Y) = \frac{1}{\sigma} \text{Log} \frac{Y}{C} - \frac{1}{\sigma} \sum_{\mathbf{n} \neq \mathbf{0}} [\mathbf{u}^{\mathbf{n}}] \mathcal{Q}_{A_{\mathbf{n}}}[H](\mathbf{u}) \prod_{j=1}^r \left(c_j (Y/C)^{-\alpha_j/\sigma} \right)^{n_j}. \quad (88)$$

The series converges whenever the vector $\mathbf{X}(Y)$ lies in a sufficiently small polydisc. Under (78), this holds for all sufficiently large positive Y ; moreover $\text{Re } A_{\mathbf{n}} > 0$ for every nonzero $\mathbf{n}$, so the first line of (87) always applies.

Proof. The analytic implicit-function theorem applies to $u_j = X_j H(\mathbf{u})^{\lambda_j}$, because the Jacobian with respect to $\mathbf{u}$ is the identity at $(\mathbf{u}, \mathbf{X}) = (\mathbf{0}, \mathbf{0})$. It remains to compute the Taylor coefficients of $z = -\log H$.

Set

$$\phi_i(\mathbf{u}) = H(\mathbf{u})^{\lambda_i}. \quad (89)$$

The r -variable Lagrange–Good formula gives

$$[\mathbf{X}^{\mathbf{n}}] z(\mathbf{u}(\mathbf{X})) = [\mathbf{u}^{\mathbf{n}}] z(\mathbf{u}) \prod_{i=1}^r \phi_i(\mathbf{u})^{n_i} \Delta(\mathbf{u}), \quad (90)$$

where

$$\Delta = \det \left(\delta_{ij} - u_j \frac{\partial}{\partial u_j} \log \phi_i \right)_{i,j=1}^r. \quad (91)$$

Since $\log \phi_i = \lambda_i \log H = -\lambda_i z$, the matrix in (91) is

$$I + \boldsymbol{\lambda} (u_1 z_{u_1}, \dots, u_r z_{u_r}). \quad (92)$$

The matrix determinant lemma therefore yields

$$\Delta = 1 + Ez, \quad E := \sum_{j=1}^r \lambda_j u_j \partial_{u_j}. \quad (93)$$

For fixed $\mathbf{n} \neq \mathbf{0}$, abbreviate $A = A_{\mathbf{n}}$ and

$$P = \prod_{i=1}^r \phi_i^{n_i} = H^A = e^{-Az}. \quad (94)$$

First suppose $A \neq 0$. For every j ,

$$u_j \partial_{u_j} \left[\left(z + \frac{1}{A} \right) P \right] = -A z u_j z_{u_j} P. \quad (95)$$

Using $[\mathbf{u}^{\mathbf{n}}]_{u_j \partial_{u_j}} Q = n_j [\mathbf{u}^{\mathbf{n}}] Q$ and summing with weights λ_j , we find

$$\begin{aligned} [\mathbf{u}^{\mathbf{n}}]_z P \Delta &= [\mathbf{u}^{\mathbf{n}}]_z P - \frac{\sum_j \lambda_j n_j}{A} [\mathbf{u}^{\mathbf{n}}] \left(z + \frac{1}{A} \right) P \\ &= -\frac{1}{A} [\mathbf{u}^{\mathbf{n}}] P = -[\mathbf{u}^{\mathbf{n}}] \mathcal{Q}_A[H]. \end{aligned} \quad (96)$$

The last equality uses $\mathbf{n} \neq \mathbf{0}$, so the constant term subtracted in $(H^A - 1)/A$ does not affect the coefficient.

If $A = 0$, then $P = 1$, and the apparent pole disappears directly:

$$\begin{aligned} [\mathbf{u}^{\mathbf{n}}]_z \Delta &= [\mathbf{u}^{\mathbf{n}}] (z + z E z) \\ &= [\mathbf{u}^{\mathbf{n}}] z + \frac{1}{2} [\mathbf{u}^{\mathbf{n}}] E(z^2) \\ &= [\mathbf{u}^{\mathbf{n}}] z + \frac{A}{2} [\mathbf{u}^{\mathbf{n}}] z^2 = -[\mathbf{u}^{\mathbf{n}}] \log H. \end{aligned} \quad (97)$$

This is exactly the $A = 0$ value in (84). Thus (86) holds in every case. Substitution in (80) proves (88). Finally, under (78), $\operatorname{Re} \lambda_j > 0$, so $X_j(Y) \rightarrow 0$ and $\operatorname{Re} A_{\mathbf{n}} > 0$ for every nonzero multi-index. $\square$

One coefficient extraction replaces reversion

Once a map is written in the form (77), the inverse coefficient indexed by $\mathbf{n}$ is obtained in three operations:

1. compute the scalar weight $A_{\mathbf{n}} = \mathbf{n} \cdot \boldsymbol{\lambda}$;
2. form the regularized quotient $\mathcal{Q}_{A_{\mathbf{n}}}[H]$;
3. extract the single coefficient $-[\mathbf{u}^{\mathbf{n}}] \mathcal{Q}_{A_{\mathbf{n}}}[H]$. When $A_{\mathbf{n}} \neq 0$, this is the familiar operation of extracting the coefficient of $H^{A_{\mathbf{n}}}$ and dividing by $-A_{\mathbf{n}}$; when $A_{\mathbf{n}} = 0$, it is $-[\mathbf{u}^{\mathbf{n}}] \log H$.

No previous inverse coefficients are needed. The triangular recurrence and Newton method remain useful computationally, but the reversion problem has been reduced to coefficient extraction in the forward perturbation.

10.2 Product and radical perturbations

The most explicit case is a product of binomial factors.

Corollary 10.2 (Product formula). *Suppose*

$$H(\mathbf{u}) = \prod_{j=1}^r (1 - u_j)^{\gamma_j}, \quad (98)$$

with arbitrary fixed complex exponents γ_j . Then

$$b_{\mathbf{n}} = \frac{(-1)^{|\mathbf{n}|+1}}{A_{\mathbf{n}}} \prod_{j=1}^r \binom{\gamma_j A_{\mathbf{n}}}{n_j}. \quad (99)$$

If $A_{\mathbf{n}} = 0$, the right-hand side is interpreted by its removable polynomial limit at zero.

Proof. Because

$$H(\mathbf{u})^A = \prod_{j=1}^r (1 - u_j)^{\gamma_j A},$$

its $\mathbf{u}^{\mathbf{n}}$ -coefficient is $(-1)^{|\mathbf{n}|} \prod_j \binom{\gamma_j A}{n_j}$. For $\mathbf{n} \neq \mathbf{0}$, this product is divisible by A as a polynomial in A , so the quotient has a removable value at zero. Apply (86) and (84). $\square$

The formula includes radicals without extra work. For example, a square root corresponds to $\gamma_j = 1/2$, and an inverse square root to $\gamma_j = -1/2$. The only branch choice is the analytic germ at $\mathbf{u} = \mathbf{0}$, where every factor equals one.

10.3 A conjugate-pair theorem for damped oscillations

A broad real family is

$$Y = C e^{\sigma x} (1 - 2r e^{-\mu x} \cos(\omega x + \delta) + r^2 e^{-2\mu x})^\gamma, \quad (100)$$

where $C, \sigma, \mu > 0$, while $r, \omega, \delta, \gamma$ are real. Factor the oscillatory polynomial as

$$\begin{aligned} 1 - 2r e^{-\mu x} \cos(\omega x + \delta) + r^2 e^{-2\mu x} \\ = \left(1 - r e^{i\delta} e^{-(\mu - i\omega)x}\right) \left(1 - r e^{-i\delta} e^{-(\mu + i\omega)x}\right). \end{aligned} \quad (101)$$

Define

$$\lambda_+ = \frac{\mu - i\omega}{\sigma}, \quad \lambda_- = \frac{\mu + i\omega}{\sigma}, \quad X_+ = r e^{i\delta} (Y/C)^{-\lambda_+}, \quad X_- = \overline{X_+} \quad (102)$$

for positive real Y/C .

Corollary 10.3 (Damped-oscillatory inverse). *For sufficiently large positive Y , the outer inverse of (100) is*

$$x(Y) = \frac{1}{\sigma} \log \frac{Y}{C} + \frac{1}{\sigma} \sum_{\substack{j,k \geq 0 \\ j+k \geq 1}} \frac{(-1)^{j+k+1}}{A_{jk}} \binom{\gamma A_{jk}}{j} \binom{\gamma A_{jk}}{k} X_+^j X_-^k, \quad (103)$$

where

$$A_{jk} = j\lambda_+ + k\lambda_-. \quad (104)$$

The series is real because its terms pair under $j \leftrightarrow k$. Grouping by $j + k$ gives powers of $(Y/C)^{-\mu/\sigma}$ with periodic coefficients in $\log(Y/C)$.

10.4 Recovery of the Fibonacci formula

For $x > 0$, equation (23) may be square rooted without ambiguity:

$$Y = \sqrt{5} \mathcal{F}(x) = e^{ax} \left(1 - e^{(-2a+i\pi)x}\right)^{1/2} \left(1 - e^{(-2a-i\pi)x}\right)^{1/2}. \quad (105)$$

Thus (77) has

$$C = 1, \quad \sigma = a, \quad \alpha_1 = 2a - i\pi, \quad \alpha_2 = 2a + i\pi, \quad \gamma_1 = \gamma_2 = \frac{1}{2}, \quad c_1 = c_2 = 1. \quad (106)$$

The general exponent ratios are

$$\frac{\alpha_1}{a} = 2 - i\frac{\pi}{a} = 2\lambda, \quad \frac{\alpha_2}{a} = 2 + i\frac{\pi}{a} = 2\bar{\lambda}, \quad (107)$$

where $\lambda = 1 - i\rho$ is the half-scaled exponent used earlier. Hence $A_{jk}^{\text{gen}} = 2A_{jk}$, and the product formula gives

$$b_{jk} = \frac{(-1)^{j+k+1}}{2A_{jk}} \binom{A_{jk}}{j} \binom{A_{jk}}{k} = \frac{1}{2} d_{jk}. \quad (108)$$

Multiplication by $1/\sigma = 1/a$ recovers the factor $d_{jk}/(2a)$ in (4). This verifies that the general universal theorem and the earlier squared-equation derivation are exactly consistent.

10.5 Elementary one-variable checks

A general formula should reproduce cases whose inverse is known without transseries machinery.

Example 10.4 (The inverse of $e^x - 1$). Take $C = \sigma = \alpha = c = 1$ and $H(u) = 1 - u$. Then $Y = e^x - 1$, so

$$x = \log(1 + Y) = \log Y + \sum_{n \geq 1} \frac{(-1)^{n+1}}{nY^n}.$$

The theorem has $A_n = n$ and

$$b_n = -\frac{1}{n} [u^n] (1 - u)^n = \frac{(-1)^{n+1}}{n},$$

which is exactly the logarithmic expansion above.

Example 10.5 (A binomially perturbed exponential). For

$$Y = e^x (1 - ce^{-\alpha x})^\gamma,$$

let $\lambda = \alpha$. The inverse correction is

$$x - \log Y = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n\lambda} \binom{\gamma n \lambda}{n} c^n Y^{-n\lambda}. \quad (109)$$

When $\gamma = 1$ and $\lambda = 1$, this reduces to the previous example. For nonintegral λ , it is a genuine complex- or fractional-power inverse series.

11 The associated transseries algebra

The analytic theorem has a formal counterpart that clarifies which operations are legitimate before convergence is considered.

11.1 The exponent semigroup

Let

$$\Gamma = \left\{ A_{\mathbf{n}} = \sum_{j=1}^r n_j \lambda_j : \mathbf{n} \in \mathbb{N}^r \right\}, \quad \text{Re } \lambda_j > 0. \quad (110)$$

The real part provides a weight

$$\text{wt}(\mathbf{X}^{\mathbf{n}}) = \text{Re } A_{\mathbf{n}}. \quad (111)$$

For every $B > 0$, only finitely many multi-indices satisfy $\text{Re } A_{\mathbf{n}} \leq B$, because

$$|\mathbf{n}| \min_j \text{Re } \lambda_j \leq \text{Re } A_{\mathbf{n}}.$$

This local finiteness makes the Cauchy product well defined coefficient by coefficient.

Definition 11.1 (Completed exponential monomial algebra). Let K be a commutative $\mathbb{Q}$ -algebra. The completed monomial algebra

$$K[[\mathbf{X}]]_{\lambda} \quad (112)$$

consists of formal sums $S = \sum_{\mathbf{n} \in \mathbb{N}^r} s_{\mathbf{n}} \mathbf{X}^{\mathbf{n}}$, ordered by the weight (111). The variables X_j are initially independent bookkeeping variables. After specialization to $X_j = c_j e^{-\lambda_j L}$, terms having the same combined exponent $A_{\mathbf{n}}$ are collected. Addition is termwise, and multiplication is the locally finite Cauchy product.

The maximal ideal $\mathfrak{m}$ consists of series with zero constant term. Every unit $1 + S$, $S \in \mathfrak{m}$, admits the usual formal operations

$$\log(1 + S), \quad (1 + S)^{\gamma}, \quad \exp(S), \quad \frac{1}{1 + S}, \quad (113)$$

formed from their ordinary Taylor series. Local finiteness guarantees that each coefficient receives only finitely many contributions.

11.2 Differentiation and complex powers

Let $L = \log(Y/C)$ and define the logarithmic derivative

$$D = \frac{d}{dL}. \quad (114)$$

Since $X_j = c_j e^{-\lambda_j L}$,

$$D \mathbf{X}^{\mathbf{n}} = -A_{\mathbf{n}} \mathbf{X}^{\mathbf{n}}. \quad (115)$$

Thus the algebra is stable under differentiation. Integration is also coefficientwise provided the constant mode is handled separately and $A_{\mathbf{n}} \neq 0$. The same scalar $A_{\mathbf{n}}$ that appears in the inverse coefficient formula is therefore the eigenvalue of $-D$ on the corresponding monomial.

When $\lambda_j = \eta_j - i\omega_j$,

$$X_j = c_j (Y/C)^{-\eta_j} e^{i\omega_j \log(Y/C)}. \quad (116)$$

The real parts η_j determine asymptotic size, while the imaginary parts ω_j generate motion on a phase torus.

Proposition 11.2 (Periodic versus quasiperiodic blocks). *Suppose the exponent set is closed under conjugation and the forward map is real on the positive real axis.*

1. *If all nonzero frequencies ω_j are integer multiples of one frequency ω_0 , the coefficient blocks are periodic in $\log(Y/C)$ with period $2\pi/\omega_0$.*
2. *If the frequencies have rational rank $s > 1$, the coefficient blocks are quasiperiodic restrictions of trigonometric polynomials on an s -torus.*
3. *Conjugation symmetry of the data implies conjugation symmetry of the inverse coefficients, and hence reality of the inverse on the physical locus.*

Proof. Every monomial phase is an integer linear combination of the basic frequencies. The first two statements are the elementary periodicity theory of such combinations. For the third, conjugate the fixed-point system and use uniqueness of its formal solution. $\square$

11.3 Formal inversion theorem

Theorem 11.3 (Formal version of [Theorem 10.1](#)). *Let $H \in 1 + (u_1, \dots, u_r)K[[\mathbf{u}]]$ and let $\lambda_j \in K$. Define $\mathcal{Q}_A[H]$ by the division-free series in (84). Then the system*

$$u_j = X_j H(\mathbf{u})^{\lambda_j}$$

has a unique solution $\mathbf{u} \in \mathfrak{m}^r$, and

$$-\log H(\mathbf{u}(\mathbf{X})) = - \sum_{\mathbf{n} \neq 0} [\mathbf{u}^{\mathbf{n}}] \mathcal{Q}_{A_{\mathbf{n}}}[H](\mathbf{u}) \mathbf{X}^{\mathbf{n}}.$$

No invertibility assumption on $A_{\mathbf{n}}$ is required.

Proof. Successive substitution determines every homogeneous part of $\mathbf{u}$ triangularly, proving existence and uniqueness. To verify the coefficient identity without hidden division, fix $\mathbf{n}$ and replace the λ_j and the finitely many coefficients of H needed through degree $|\mathbf{n}|$ by algebraically independent indeterminates. Over the fraction field of the resulting universal polynomial ring, the nonzero linear form $A_{\mathbf{n}} = \sum_j n_j \lambda_j$ is invertible, so the algebraic Lagrange–Good proof of [Theorem 10.1](#) applies. On the other hand,

$$[\mathbf{u}^{\mathbf{n}}] \mathcal{Q}_A[H] = \sum_{m=1}^{|\mathbf{n}|} \frac{A_{\mathbf{n}}^{m-1}}{m!} [\mathbf{u}^{\mathbf{n}}] (\log H)^m$$

is polynomial in all of those indeterminates. Hence the identity descends from the fraction field to the universal polynomial ring and may then be specialized to arbitrary coefficients of H and arbitrary $\lambda_j \in K$, including specializations for which some $A_{\mathbf{n}}$ vanishes or is not a unit. $\square$

This result separates two issues that are often conflated. Formal inversion requires local finiteness, but after regularization it requires no division by a combined exponent. Analytic convergence additionally requires bounds on the forward germ and a domain in which the implicit solution remains regular.

11.4 Zero combined exponents and failure of the chosen scale

The factor $A_{\mathbf{n}}^{-1}$ in the convenient nonzero-weight formula looks like a resonance denominator, but in this common-factor inversion problem its pole is always removable. Indeed,

$$\frac{H^A - 1}{A} = \log H + \frac{A}{2!} (\log H)^2 + \frac{A^2}{3!} (\log H)^3 + \dots, \quad (117)$$

so a zero combined exponent contributes $-[\mathbf{u}^{\mathbf{n}}] \log H$, not a logarithmic factor in Y .

What may fail is the chosen *asymptotic ordering*.

1. A monomial with $\operatorname{Re} \lambda_j = 0$ does not decay. It normally belongs in the leading problem rather than in a perturbative tail.
2. If some $\operatorname{Re} \lambda_j < 0$, the corresponding supposed small variable grows, so the chart is invalid in that direction.
3. Distinct multi-indices may have the same exponent $A_{\mathbf{n}}$ after one-variable specialization. Their coefficients must then be collected; if the common exponent is zero, the result is a constant-scale contribution that should usually be absorbed into the leading balance.

Logarithmic monomials can certainly arise in broader transseries problems—for example when a differential equation requires inversion of D on a zero-eigenvalue forcing, or when the dominant envelope already contains powers of x and $\log x$. They are not forced merely by $A_{\mathbf{n}} = 0$ in the analytic common-factor reversion treated here.

12 Extensions of the method

12.1 Several damped harmonics

A finite exponential–trigonometric perturbation can always be written using complex exponentials. For example,

$$Y = Ce^{\sigma x} H\left(r_1 e^{-(\mu_1 - i\omega_1)x}, r_1 e^{-(\mu_1 + i\omega_1)x}, \dots, r_s e^{-(\mu_s + i\omega_s)x}\right) \quad (118)$$

falls directly under [Theorem 10.1](#). If the ω_j/σ are commensurable, one obtains periodic coefficient blocks. Otherwise the inverse contains quasiperiodic coefficient functions. The universal formula remains a finite coefficient extraction at every multi-index.

The damping rates need not be equal. In that case total degree is no longer the most efficient truncation rule. One should order monomials by

$$\text{wt}(\mathbf{n}) = \sum_j n_j \frac{\mu_j}{\sigma}, \quad (119)$$

and retain all terms below a prescribed weight. Local finiteness still holds because every $\mu_j > 0$.

12.2 A polynomial or logarithmic envelope

The exact closed formula relies on a pure exponential envelope, for which a constant displacement z/σ multiplies the envelope by e^z . A more general map may have the form

$$Y = M(x) H(c_1 e^{-\alpha_1 x}, \dots, c_r e^{-\alpha_r x}), \quad (120)$$

where, for example,

$$M(x) = Ce^{\sigma x} x^\beta (\log x)^\gamma. \quad (121)$$

The method then proceeds in two layers.

First invert the dominant envelope:

$$x_0 = M^{-1}(Y). \quad (122)$$

For (121), this base inverse is expressed through Lambert-type or polynomial–logarithmic transseries. Next write $x = x_0 + \delta$. The exact correction equation is

$$0 = \log \frac{M(x_0 + \delta)}{M(x_0)} + \log H\left(c_j e^{-\alpha_j(x_0 + \delta)}\right). \quad (123)$$

Its derivative at $\delta = 0$ is asymptotic to $(\log M)'(x_0)$, which is nonzero in the outer regime. The analytic implicit-function theorem and one-variable Lagrange–Bürmann inversion then produce δ in a coefficient algebra whose elements are slow transseries in x_0 . The first correction is universally

$$\delta = -\frac{\log H(c_j e^{-\alpha_j x_0})}{(\log M)'(x_0)} + \text{higher perturbative orders}. \quad (124)$$

Thus the pure exponential theorem is the constant-slope member of a broader anchored reversion calculus.

12.3 Lagrange–Bürmann over a coefficient algebra

Suppose an anchored correction can be written as

$$\delta = \varepsilon \Phi(\delta), \quad (125)$$

where ε is a small monomial and the Taylor coefficients of Φ belong to a differential algebra of slower asymptotic functions. Then formally

$$\delta = \sum_{n \geq 1} \frac{\varepsilon^n}{n!} \frac{\partial^{n-1}}{\partial s^{n-1}} \Phi(s)^n \Big|_{s=0}. \quad (126)$$

The formula is ordinary Lagrange–Bürmann inversion, but interpreted over the slow coefficient algebra. This observation connects the present complex-power theory with polynomial–logarithmic transseries: one scale can serve as the coefficient field for reversion in a smaller scale.

12.4 Complex branches

For complex Y , every choice in

$$L = \text{Log}(Y/C), \quad H(\mathbf{u})^{\lambda_j} = \exp(\lambda_j \text{Log } H(\mathbf{u})) \quad (127)$$

selects an asymptotic inverse chart. The coefficient formula is unchanged; only the monomial phases and analytic continuation paths differ. Singular curves arise where the Lagrange–Good Jacobian vanishes or where the selected branch of H becomes singular. Analytic continuation around such curves can permute inverse branches.

For the Fibonacci continuation, squaring removes the outer square root but also merges sign information. On the positive real branch this is harmless, because $\mathcal{F}(x) > 0$ for $x > 0$. In the complex plane, however, one must specify whether one is inverting the squared map, the two-factor square-root map, or a chosen complex Binet continuation. Their Riemann surfaces are related but not identical.

12.5 Other linear recurrences

Let a second-order recurrence have characteristic roots r_1, r_2 with $|r_1| > |r_2|$. A Binet-type interpolation has the form

$$U(x) = Ar_1^x + Br_2^x. \quad (128)$$

After factoring the dominant term,

$$U(x) = Ar_1^x \left(1 + \frac{B}{A} e^{-\alpha x} \right), \quad \alpha = \text{Log}(r_1/r_2), \quad (129)$$

so its inverse is a one-factor instance of [Theorem 10.1](#). If a modulus or quadratic norm is taken, a conjugate pair appears and the inverse becomes log-periodic whenever α has nonzero imaginary part. Higher-order recurrences lead to several small root ratios and hence to multivariate, often quasiperiodic, transseries.

13 Further consequences for the Fibonacci inverse

13.1 Sharp first-order oscillation

The first block gives more than an $O(y^{-2})$ estimate.

Corollary 13.1 (No limiting normalized coefficient). *As $y \rightarrow +\infty$,*

$$5ay^2 \left(\Psi(y) - \frac{1}{a} \log(\sqrt{5}y) \right) = \cos \left(\frac{\pi}{a} \log(\sqrt{5}y) \right) + O(y^{-2}). \quad (130)$$

Consequently

$$\limsup_{y \rightarrow \infty} 5ay^2 \left(\Psi(y) - \frac{1}{a} \log(\sqrt{5}y) \right) = 1, \quad (131)$$

$$\liminf_{y \rightarrow \infty} 5ay^2 \left(\Psi(y) - \frac{1}{a} \log(\sqrt{5}y) \right) = -1. \quad (132)$$

In particular, the normalized first correction has no limit.

Proof. Equation (130) is (5) with the first block retained. The phase $(\pi/a) \log(\sqrt{5}y)$ increases continuously to infinity, so there are sequences of y 's on which its cosine is respectively 1 and -1 . The error tends to zero along both sequences. $\square$

13.2 Ordinary series along geometric phase rays

Fix $Y_0 > 0$ and define

$$Y_m = Y_0 \varphi^{2m}, \quad y_m = Y_m / \sqrt{5}. \quad (133)$$

Then

$$\theta_m = \frac{\pi}{a} \log Y_m = \frac{\pi}{a} \log Y_0 + 2\pi m, \quad (134)$$

so every trigonometric block has a fixed coefficient. Along this geometric sequence,

$$\Psi(y_m) = \frac{\log Y_m}{a} + \frac{1}{2a} \sum_{n \geq 1} Y_m^{-2n} Z_n(\theta_0), \quad (135)$$

which is an ordinary convergent power series in Y_m^{-2} . Thus the full log-periodic transseries may be viewed as a continuous family of ordinary power series indexed by a phase on the circle.

13.3 A phase-averaged elliptic generating function

Let

$$\bar{z}(t) = \frac{1}{2\pi} \int_0^{2\pi} z(t, \theta) \, d\theta. \quad (136)$$

Using (58),

$$\bar{z}(t) = -\frac{1}{2} \sum_{m \geq 1} \frac{1}{m} \binom{2m}{m} t^{2m}. \quad (137)$$

Put $s = 16t^2$. Since

$$\binom{2m}{m} = 4^m \frac{(1/2)_m}{m!}, \quad (138)$$

one obtains

$$s \frac{d\bar{z}}{ds} = -\frac{1}{2} \left({}_2F_1 \left(\frac{1}{2}, \frac{1}{2}; 1; s \right) - 1 \right). \quad (139)$$

With $K(s)$ denoting the complete elliptic integral of the first kind in the parameter convention

$$K(s) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - s \sin^2 \phi}}, \quad (140)$$

we have ${}_2F_1(1/2, 1/2; 1; s) = 2K(s)/\pi$, and therefore

$$\boxed{\bar{z}(t) = \int_0^{16t^2} \left(\frac{1}{2} - \frac{K(s)}{\pi} \right) \frac{ds}{s}.} \quad (141)$$

The logarithmic singularity of $K(s)$ at $s = 1$ is consistent with the radius $t = 1/4$ of the averaged subseries. This elliptic connection arises solely from the diagonal Lagrange–Good coefficients.

13.4 All logarithmic derivatives

Let

$$D_y = y \frac{d}{dy} = \frac{d}{d \log Y}. \quad (142)$$

Because $D_y(X^j \bar{X}^k) = -2A_{jk}X^j \bar{X}^k$, termwise differentiation of the convergent series gives, for $m \geq 1$,

$$D_y^m \Psi(y) = \frac{\delta_{m1}}{a} + \frac{1}{2a} \sum_{j+k \geq 1} d_{jk} (-2A_{jk})^m X^j \bar{X}^k. \quad (143)$$

Equivalently, in block form,

$$D_y \Psi(y) = \frac{1}{a} + \frac{1}{2a} \sum_{n \geq 1} t^n \left(-2n Z_n(\theta) + \frac{\pi}{a} Z'_n(\theta) \right). \quad (144)$$

The leading correction is

$$y \Psi'(y) = \frac{1}{a} - \frac{2t}{a} \cos \theta - \frac{\pi t}{a^2} \sin \theta + O(t^2). \quad (145)$$

Higher ordinary derivatives follow by converting powers of D_y through Stirling-number identities. Thus the same coefficient table controls the inverse and all of its derivatives.

13.5 Identities on the Fibonacci lattice

For every integer $n \geq 2$, $\Psi(F_n) = n$. Therefore the convergent transseries supplies the identity

$$n = \frac{1}{a} \log(\sqrt{5}F_n) + \frac{1}{2a} \sum_{j+k \geq 1} d_{jk} X_n^j \bar{X}_n^k, \quad (146)$$

where

$$X_n = (\sqrt{5}F_n)^{-2+i\pi/a}. \quad (147)$$

For all sufficiently large n , the series is convergent; its truncations provide rapidly improving corrections to the elementary approximation $n \approx a^{-1} \log(\sqrt{5}F_n)$. Binet's exact identity

$$\sqrt{5}F_n = \varphi^n (1 - (-1)^n \varphi^{-2n}) \quad (148)$$

shows that (146) is ultimately an intricate re-expansion of a simple logarithm. The value of the transseries formula is that it remains valid for every large real y , not only on the integer lattice.

14 Computer algebra and numerical implementation

This section gives self-contained code for generating the blocks and checking them numerically. The code is included in the article rather than required as an external build dependency.

14.1 Wolfram Language: triangular block generator

The following program implements (63). It is robust for symbolic generation because each step expands only to the order currently needed.

Listing 1: Generating the trigonometric blocks Z_n .

```

ClearAll[t, th, rho, inverseBlocks];

inverseBlocks[nmax_Integer?Positive] := Module[
  {zser = 0, blocks = {}, n, residual, zn},
  Do[
    residual = Normal @ Series[
      Exp[zser] + t^2 Exp[-zser]
      - 2 t Cos[th + rho zser] - 1,
      {t, 0, n}
    ];
    zn = FullSimplify[
      TrigReduce[-Coefficient[residual, t, n]],
      Assumptions -> Element[{rho, th}, Reals]
    ];
    AppendTo[blocks, zn];
    zser = zser + zn t^n,
    {n, 1, nmax}
  ];
  blocks
];

blocks = inverseBlocks[6];
TableForm[MapIndexed[{First[#2], #1} &, blocks]]

```

A direct implementation of the closed formula is shorter:

Listing 2: Closed Lagrange–Good coefficients.

```

ClearAll[aa, d, zBlock, rho, th];

aa[j_, k_] := j (1 - I rho) + k (1 + I rho);

d[j_, k_] := (-1)^(j + k + 1)
  Binomial[aa[j, k], j] Binomial[aa[j, k], k] / aa[j, k];

zBlock[n_Integer?Positive] := FullSimplify[
  ComplexExpand[
    Sum[d[j, n - j] Exp[I (2 j - n) th], {j, 0, n}],
    TargetFunctions -> {Re, Im}
  ],
  Assumptions -> Element[{rho, th}, Reals]
];

Table[zBlock[n], {n, 1, 6}]

```

14.2 Wolfram Language: inverse approximation and residual

Listing 3: Evaluating and certifying a truncation.

```

ClearAll[a, rhoValue, fibCont, psiApprox, residual];

a = ArcCsch[2];
rhoValue = Pi/(2 a);

fibCont[x_?NumericQ] := Sqrt[2/5]
  Sqrt[Cosh[2 a x] - Cos[Pi x]];

```

```

psiApprox[y_?NumericQ, nmax_Integer?NonNegative] := Module[
  {Y, tt, phase, corr},
  Y = Sqrt[5] y;
  tt = Y^(-2);
  phase = (Pi/a) Log[Y];
  corr = Sum[
    tt^n (zBlock[n] /. {rho -> rhoValue, th -> phase}),
    {n, 1, nmax}
  ];
  Log[Y]/a + corr/(2 a)
];

residual[y_?NumericQ, nmax_Integer?NonNegative] := Module[
  {Y, tt, phase, z0},
  Y = Sqrt[5] y;
  tt = Y^(-2);
  phase = (Pi/a) Log[Y];
  z0 = 2 a (psiApprox[y, nmax] - Log[Y]/a);
  Exp[z0] + tt^2 Exp[-z0]
  - 2 tt Cos[phase + rhoValue z0] - 1
];

(* Independent direct inversion on the outer branch *)
psiDirect[y_?NumericQ] := x /. FindRoot[
  fibCont[x] == y,
  {x, Log[Sqrt[5] y]/a},
  WorkingPrecision -> 80
];

```

14.3 Python/mpmath numerical checker

Listing 4: High-precision numerical comparison with the defining equation.

```

import mpmath as mp

mp.mp.dps = 80
a = mp.asinh(mp.mpf("0.5"))
rho = mp.pi / (2 * a)

def fib_cont(x: mp.mpf) -> mp.mpf:
    """The real Binet-Fibonacci continuation."""
    return mp.sqrt(mp.mpf(2) / 5) * mp.sqrt(
        mp.cosh(2 * a * x) - mp.cos(mp.pi * x)
    )

def generalized_binomial(z: complex, n: int) -> complex:
    """Binomial(z,n) for a nonnegative integer n."""
    value = mp.mpf(1)
    for m in range(n):
        value *= z - m
    return value / mp.factorial(n)

```

```

def z_block(n: int, theta: mp.mpf) -> mp.mpf:
    """Evaluate Z_n(theta) from the closed coefficient formula."""
    lam = 1 - 1j * rho
    total = 0j
    for j in range(n + 1):
        k = n - j
        A = j * lam + k * lam.conjugate()
        d = ((-1) ** (n + 1) / A
              * generalized_binomial(A, j)
              * generalized_binomial(A, k))
        total += d * mp.e ** (1j * (2 * j - n) * theta)
    return mp.re(total)

def psi_approx(y: mp.mpf, order: int) -> mp.mpf:
    """N-block inverse transseries approximation."""
    Y = mp.sqrt(5) * y
    t = Y ** (-2)
    theta = (mp.pi / a) * mp.log(Y)
    z = mp.fsum(t ** n * z_block(n, theta)
                for n in range(1, order + 1))
    return mp.log(Y) / a + z / (2 * a)

def psi_direct(y: mp.mpf) -> mp.mpf:
    """Independent high-precision root on the outer positive branch."""
    x0 = mp.log(mp.sqrt(5) * y) / a
    return mp.findroot(lambda x: fib_cont(x) - y,
                      (x0 - mp.mpf("0.2"), x0 + mp.mpf("0.2")))

for y in map(mp.mpf, [2, 5, 10, 50, 100]):
    exact = psi_direct(y)
    errors = [abs(psi_approx(y, n) - exact)
              for n in [0, 1, 2, 4, 6]]
    print(y, exact, errors)

```

14.4 Implementation cautions

Three small details prevent common errors.

1. Use the same branch of Log in t^λ and in the phase θ . For positive y , the real logarithm is canonical.
2. Group conjugate coefficient pairs before converting to machine precision. This suppresses spurious imaginary roundoff.
3. Near the convergence boundary, evaluate a moderate truncation and then apply one or two Newton steps to the exact equation, rather than summing many large, canceling blocks.

15 Research directions and open problems

The closed coefficient formula resolves the all-orders large-argument inversion, but it also exposes several natural frontier questions.

15.1 The exact convergence boundary

The analytic solution in the independent variables (X, V) can fail only when a selected binomial branch becomes singular or when the Lagrange–Good Jacobian vanishes. For the Fibonacci system,

$$\Delta(u, v) = 1 + \frac{\lambda u}{1 - u} + \frac{\bar{\lambda} v}{1 - v}. \quad (149)$$

A complete convergence analysis therefore begins with the singular system

$$\begin{cases} u = X(1 - u)^\lambda(1 - v)^\lambda, \\ v = V(1 - u)^{\bar{\lambda}}(1 - v)^{\bar{\lambda}}, \\ \Delta(u, v) = 0, \end{cases} \quad (150)$$

together with the branch divisors $u = 1$ and $v = 1$. The problem is to locate the singularity of smallest modulus in each physical direction $(X, V) = (te^{i\theta}, te^{-i\theta})$, and then minimize over θ . This would replace the qualitative phrase “sufficiently large” by a sharp convergence threshold.

Conjecture 15.1 (Generic fold singularities on fixed phase rays). Fix a real phase θ and restrict the bivariate solution to $(X, V) = (se^{i\theta}, se^{-i\theta})$. Except possibly for a discrete set of phases, the dominant singularities of the resulting one-variable function $z_\theta(s)$ are simple folds: they satisfy (150), avoid $u = 1$ and $v = 1$, and have a nonzero second derivative in the null direction. Consequently the large-order coefficients of z_θ are finite sums of square-root singularity contributions of the form $CR^{-n}n^{-3/2}$, possibly with oscillatory factors from conjugate singularities.

This is the generic behavior predicted by analytic implicit-function theory, but a proof requires global control of the singular variety. Multivariate analytic-combinatorics methods may be useful for the homogeneous coefficient sums; see [2] for the univariate background.

15.2 Large-order asymptotics of the Fourier blocks

The exact formula (6) reduces the problem to a finite sum of complex generalized binomial coefficients. Several regimes are of interest:

1. fixed Fourier mode $m = 2j - n$ as $n \rightarrow \infty$;
2. bulk scaling $j/n \rightarrow \xi \in (0, 1)$;
3. edge scaling j fixed or $n - j$ fixed;
4. maximum norms $\max_\theta |Z_n(\theta)|$.

Stirling analysis of the gamma-function representation

$$\binom{A}{j} = \frac{\Gamma(A + 1)}{\Gamma(j + 1)\Gamma(A - j + 1)} \quad (151)$$

should identify the dominant Fourier modes and connect coefficient growth to the singular geometry of (150).

15.3 Uniform approximations near the convergence boundary

Near a fold singularity, summing the Taylor transseries is not the best representation. One should introduce a local square-root variable and build a uniform approximant that matches the large-argument series away from the fold. Rational, Padé, Hermite–Padé, or conformally mapped approximants could then extend accurate evaluation toward $y = 1$. The exact residual certificate in Theorem 7.3 provides a convenient way to validate such accelerations.

15.4 All real and complex inverse branches

The outer branches are the only branches that survive for large real values, but the bounded oscillatory region contains additional local inverses. At a nondegenerate critical point, these are Puiseux series in $(y - y_c)^{1/2}$; [Section A](#) records the leading terms. A complete real branch atlas would locate every critical point and critical value. The complex problem is richer: branch points of the square root in $\mathcal{F}$, critical points of the complexified map, and logarithmic monodromy of the inverse all interact.

15.5 Other recurrences and exponential sums

Theorem [10.1](#) applies immediately to Binet formulas for Lucas sequences and, after choosing a dominant root, to higher-order linear recurrences. A systematic classification could answer:

1. which root configurations yield ordinary powers, log-periodic blocks, or quasiperiodic blocks;
2. how repeated characteristic roots introduce polynomial factors and therefore nested polynomial–logarithmic scales;
3. how moduli, norms, and real-part projections alter branch structure;
4. whether arithmetic sampling at integer arguments forces additional identities among the coefficient blocks.

15.6 Formalization

A proof-assistant development could be organized in modular layers:

1. formal multivariate power series and generalized binomial powers;
2. the Lagrange–Good coefficient identity;
3. the rank-one determinant lemma and cancellation [\(96\)](#);
4. analytic implicit-function and convergence transfer;
5. the Fibonacci specialization and explicit first blocks;
6. interval-certified residual bounds for numerical examples.

The universal theorem is especially suitable for formalization because the hard-looking inversion collapses to a short algebraic identity once the standard multivariate inversion theorem is available.

15.7 Interaction with broader transseries scales

The exponential monomial algebra here can be nested inside slower scales, as in [Section 12](#). Conversely, the coefficient blocks may themselves carry q -series, special-function, or recursively defined asymptotics. Developing a systematic tower

$$\text{polynomial–logarithmic scale} \supset \text{exponential small scale} \supset \text{log-periodic or quasiperiodic blocks} \quad (152)$$

would connect the present note with the wider algebraic theory of transseries [\[9, 1\]](#). The essential design principle is always the same: choose an ordered, locally finite monomial family and perform reversion over the coefficient algebra generated by slower scales.

16 Conclusion

The real continuation

$$\mathcal{F}(x) = \sqrt{\frac{2}{5}} \sqrt{\cosh(2x \log \varphi) - \cos(\pi x)}$$

looks elementary, but its inverse combines three distinct mechanisms: exponential growth, exponentially damped oscillation, and nonlinear phase feedback. The dominant envelope gives the logarithm $a^{-1} \log(\sqrt{5}y)$. The damped oscillation survives inversion as a log-periodic hierarchy in $(5y^2)^{-1}$. Nonlinear feedback creates higher harmonics and sine components.

The decisive exact reduction is

$$u = X(1-u)^\lambda(1-v)^\lambda, \quad v = V(1-u)^{\bar{\lambda}}(1-v)^{\bar{\lambda}}, \quad z = -\log((1-u)(1-v)).$$

Multivariate Lagrange–Good inversion then yields the explicit coefficient

$$d_{jk} = \frac{(-1)^{j+k+1}}{j\lambda + k\bar{\lambda}} \binom{j\lambda + k\bar{\lambda}}{j} \binom{j\lambda + k\bar{\lambda}}{k}.$$

This formula proves the all-orders expansion, its reality, Fourier support, parity, and convergence for large arguments. It also gives a practical algorithm and explains the unexpectedly rapid numerical accuracy.

More importantly, the example reveals a universal principle. For every map

$$Y = Ce^{\sigma x} H(c_j e^{-\alpha_j x}),$$

the inverse coefficient indexed by $\mathbf{n}$ is simply

$$-\frac{1}{A_{\mathbf{n}}} [\mathbf{u}^{\mathbf{n}}] H(\mathbf{u})^{A_{\mathbf{n}}}, \quad A_{\mathbf{n}} = \sum_j n_j \frac{\alpha_j}{\sigma}.$$

The entire reversion is encoded in ordinary coefficients of scalar powers of the forward perturbation. Complex-conjugate exponent ratios produce log-periodic blocks; several independent imaginary parts produce quasiperiodicity; positive real parts guarantee a decaying, locally finite small scale. Even outside that regime, the apparent zero-weight poles in the coefficient formula are removable.

Thus the inverse Fibonacci continuation is not an isolated special-function calculation. It is a model example of a broad and explicitly solvable class of exponential–oscillatory inverse problems.

A Local inverse branches at finite critical values

For completeness, we record the generic local behavior of the additional inverse branches that occur in the bounded oscillatory region. This is separate from the large-argument transseries.

Proposition A.1 (Puiseux inversion at a simple critical point). *Let $x_c \neq 0$ satisfy*

$$\mathcal{F}'(x_c) = 0, \quad \mathcal{F}''(x_c) \neq 0, \quad y_c = \mathcal{F}(x_c). \quad (153)$$

Then the two local inverse branches have the Puiseux expansions

$$x = x_c \pm \sqrt{\frac{2(y - y_c)}{\mathcal{F}''(x_c)}} - \frac{\mathcal{F}'''(x_c)}{3\mathcal{F}''(x_c)^2} (y - y_c) + O((y - y_c)^{3/2}), \quad (154)$$

with the square root interpreted in the appropriate real or complex sector.

Proof. Write $h = x - x_c$ and $s = y - y_c$. Taylor expansion gives

$$s = Ah^2 + Bh^3 + O(h^4), \quad A = \frac{1}{2}\mathcal{F}''(x_c), \quad B = \frac{1}{6}\mathcal{F}'''(x_c).$$

Seek $h = c_1 s^{1/2} + c_2 s + O(s^{3/2})$. Matching the coefficients of s and $s^{3/2}$ yields

$$Ac_1^2 = 1, \quad 2Ac_1c_2 + Bc_1^3 = 0.$$

Thus $c_1 = \pm A^{-1/2}$ and $c_2 = -B/(2A^2) = -\mathcal{F}'''(x_c)/(3\mathcal{F}''(x_c)^2)$, proving the formula. $\square$

This square-root branching is qualitatively different from the outer inverse at infinity, whose correction is analytic in the complex small monomials X and V .

B Coefficient table in complex form

For low degrees, it is sometimes more compact to retain the complex coefficients rather than expand into sines and cosines. Let $A_{jk} = j + k - i\rho(j - k)$. The homogeneous block of degree n is

$$Z_n(\theta) = \sum_{j=0}^n d_{j,n-j} e^{i(2j-n)\theta}, \quad d_{jk} = \frac{(-1)^{j+k+1}}{A_{jk}} \binom{A_{jk}}{j} \binom{A_{jk}}{k}. \quad (155)$$

The boundary coefficients simplify to

$$d_{n0} = \frac{(-1)^{n+1}}{n\lambda} \binom{n\lambda}{n}, \quad d_{0n} = \overline{d_{n0}}, \quad (156)$$

and, for even degree $2m$, the diagonal coefficient is

$$d_{mm} = -\frac{1}{2m} \binom{2m}{m}^2. \quad (157)$$

These two families control respectively the highest Fourier harmonics and the phase averages.

C Notation summary

Symbol	Meaning
φ	Golden ratio $(1 + \sqrt{5})/2$.
a	$\log \varphi = \operatorname{arsinh}(1/2) = \operatorname{arccsch} 2$.
$\mathcal{F}(x)$	Real nonnegative Fibonacci continuation in the problem statement.
$\Psi(y)$	Positive outer inverse, valued in $[2, \infty)$.
Y	Scaled argument $\sqrt{5}y$.
t	Small real parameter $Y^{-2} = (5y^2)^{-1}$.
θ	Logarithmic phase $(\pi/a) \log Y$.
ρ	Phase-feedback constant $\pi/(2a)$.
z	Dimensionless inverse displacement defined by $\Psi = a^{-1} \log Y + z/(2a)$.
λ	Complex exponent $1 - i\rho$.
X, V	Independent small monomials $t^\lambda, t^{\bar{\lambda}}$; on the physical locus $V = \bar{X}$.
A_{jk}	Combined exponent $j\lambda + k\bar{\lambda}$.
d_{jk}	Closed double-series coefficient in (40).
$Z_n(\theta)$	Real trigonometric coefficient block of order t^n .
$H(z; t, \theta)$	Exact scalar residual in (21).
$H(\mathbf{u})$	General analytic perturbation germ in Section 10; context distinguishes it from the scalar residual.
A_n	General combined exponent $\sum_j n_j \alpha_j / \sigma$.
$\mathcal{Q}_A[H]$	Regularized power quotient $(H^A - 1)/A$, continued at $A = 0$ by $\log H$.
b_n	Universal inverse coefficient $-[\mathbf{u}^n] \mathcal{Q}_{A_n}[H]$; for $A_n \neq 0$, this is $-A_n^{-1}[\mathbf{u}^n] H^{A_n}$.

D A compact derivation checklist

For a new exponential–oscillatory inverse problem, the following checklist usually leads directly to the answer.

1. Factor the forward map as $Ce^{\sigma x}H(\mathbf{u})$ with $u_j = c_j e^{-\alpha_j x}$ and $H(0) = 1$.
2. Choose $L = \text{Log}(Y/C)$, $x = (L + z)/\sigma$, and $X_j = c_j e^{-(\alpha_j/\sigma)L}$.
3. Verify that $\text{Re}(\alpha_j/\sigma) > 0$ in the asymptotic sector.
4. Form $A_n = \sum_j n_j \alpha_j / \sigma$.
5. Extract

$$b_n = -[\mathbf{u}^n] \mathcal{Q}_{A_n}[H].$$

For $A_n \neq 0$, this is $-A_n^{-1}[\mathbf{u}^n]H^{A_n}$; for $A_n = 0$, use $-\mathbf{u}^n \log H$.

6. Pair conjugate indices and group equal real weights into periodic or quasiperiodic blocks.
7. Check the first blocks by substitution into the exact normalized equation.
8. For numerical use, evaluate a moderate truncation, compute its exact residual, and finish with Newton iteration plus a derivative bound.

References

- [1] O. Costin, *Asymptotics and Borel Summability*, Chapman & Hall/CRC Monographs and Surveys in Pure and Applied Mathematics, vol. 141, CRC Press, 2009.
- [2] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [3] I. M. Gessel, A combinatorial proof of the multivariable Lagrange inversion formula, *Journal of Combinatorial Theory, Series A* **45** (1987), 178–195.
- [4] I. M. Gessel, Lagrange inversion, *Journal of Combinatorial Theory, Series A* **144** (2016), 212–249.
- [5] I. J. Good, Generalizations to several variables of Lagrange’s expansion, with applications to stochastic processes, *Proceedings of the Cambridge Philosophical Society* **56** (1960), 367–380.
- [6] M. Özvatan and O. K. Pashaev, Generalized Fibonacci sequences and Binet–Fibonacci curves, *arXiv:1707.09151* (2017).
- [7] V. Reshetnikov, *ProveIt: Series and Transseries Drafts*, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>, accessed 3 September 2026.
- [8] A. D. Sokal, A ridiculously simple and explicit implicit function theorem, *Séminaire Lotharingien de Combinatoire* **61A** (2009), Article B61Ad, 21 pp.
- [9] J. van der Hoeven, *Transseries and Real Differential Algebra*, Lecture Notes in Mathematics, vol. 1888, Springer, 2006.